TSN’s Application in Industrial Cyber Physical Systems

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Outline

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Problem Statement

- Safety-critical CPS (industrial automation, robotics, vehicles) require deterministic, low-latency communication
- Wired TSN provides bounded latency & microsecond-level jitter control
- Wireless determinism remains unachieved in practice
- Wireless challenges: interference, fading, congestion, mobility
- Precise synchronization & admission control become difficult
- Uncertainty directly degrades control loops and robotic motion
- Admission control and network planning are critical for scalable deployment
- Over 6 million vehicle accidents annually in the U.S. highlight the need for reliable CPS networking

Why Time Sensitive Networking (TSN) Matters?

Supports real-time CPS via:

- Time-aware scheduling (IEEE 802.1Qbv)
- Stream reservation (IEEE 802.1Qcc)
- Precise time sync (IEEE 802.1AS)

Enables predictable, bounded, collision-free communication

Essential for motion control, automation cells, safety systems

Introduction

- Industry 4.0 drives need for networked sensors, robots, and vehicles
- Deterministic communication is required—not optional
- Wired TSN is mature; wireless TSN is still developing
- Wireless introduces:
  - Spectrum congestion
  - Channel fading
  - Variable delays
  - Interference

Research example: **PkfR (Meng & Zhang)** demonstrates URLLC reliability via joint scheduling + power control in 5G networks

Cyber-Physical Systems Networking Problems

CPS requires:

- Bounded latency
- Low jitter
- High reliability

Wireless CPS issues:

- Variable access delays
- Mobility & multi-hop challenges
- Interference from co-located devices
- Timing accumulation & jitter over hops

TSN mechanisms designed for collision-free Ethernet, not wireless

Existing TSN Solutions

Scheduling

- IEEE 802.1Qcc
- Stream reservation
- Streams get bandwidth and timely delivery

Synchronization

- IEEE 802.1AS Time Synchronization
- Common Time Reference

Traffic Shaping

- IEEE 802.1Qbv
- TDMA based division of time slots to dedicated classes
- When to open and close gates

Current TSN Applications

Robotics:

TSN provides low latency and packet loss that is ultra beneficial for robotics applications.

- Queuing delay in ethernet switches lack proper isolation of priority traffic
- TSN Time-Aware shaping provides **priority traffic isolation, low latency, and low jitter** even with high bandwidth background traffic present [5]

Current TSN Applications

Automotive:

Through the **synchronization** of networked sensors within a vehicle or V2V system, vehicles are/will be capable of:

- Infrastructure/environment awareness (traffic lights, road conditions, weather)
- Collision avoidance and emergency braking
- Increased fuel efficiency/fuel economy control

Precise time synchronization enables deterministic behavior in automotive systems, where timely and accurate data is critical to operator safety.

Current TSN Challenges

The operating environments for robotic, automation, and automotive technologies is **inherently dynamic**.

- Factories with other networked and non-networked machinery
- High vehicle and pedestrian traffic areas
- City infrastructure (buildings, tunnels, overpasses etc.)
- Remote areas with natural obstacles like forests or mountains

Varying levels of interference are produced by different environments. TSN solutions need to be **scalable** and **robust** to handle variance and produce deterministic results.

Open Problems in Applications of TSNs

Robotics:

- Time-Aware Scheduler (TAS) based on a TDMA defined in 802.1Qbv
- Guarantees access to the MAC is made without delay in time critical applications

Automotive:

- IEEE 802.11p standard which utilizes CSMA/CA to coordinate access and interference control
- ADHOC MAC protocol (AMAC) and VDDCP [2]

Open Problems

- MAC protocols in TSN
  - End-nodes difficulty learning MAC addresses efficiently
  - Interoperability at the MAC and physical layers
- Time Synchronization across all devices
  - Limited fault tolerance through Best Master Clock Algorithm (BMCA)
- Open and closing gates in IEEE 802.1Qbv
  - Determining optimal schedule given network constraints is NP-hard

Open Problems

- Priority within a class in IEEE 802.1Qbu
  - Only two traffic classes, express and preemptable
  - Lacks flexibility- frames in same category can’t preempt each other
- Network Reliability
  - Relies on time and space redundancy for fault handling
  - Limited guidance, scheduling complexity, and high bandwidth hinder efficient deployment
- Time as an attack vector
  - Delays can be induced by an attacker

Promising Solution Approaches

- MAC protocols in TSN
  - Limit TSN bridges to one globally unique MAC address for end-nodes
  - Real-time capable firewalls
- Time Synchronization across all devices
  - IEEE 802.1ASdm introduces hot standby for redundant synchronization paths
- Open and closing gates in IEEE 802.1Qbv
  - Integer Linear Programming-finds the optimal Gate Control List

Promising Solution Approaches

- Priority within a class in IEEE 802.1Qbu
  - Multi-level preemption
  - More than two priority classes
- Network Reliability
  - Traffic redundancy using HSR-ware TSN
  - Standardized implementation profiles or reference designs
- Time as an attack vector
  - Machine learning-based intrusion detection framework

Conclusion

- Deterministic communication is essential for safety-critical cyber-physical systems.
- TSN brings bounded latency, low jitter, and reliable synchronization to industrial, robotic, and automotive networks.
- Wireless TSN is the biggest remaining challenge due to interference, mobility, and timing instability.
- Current TSN standards (802.1Qbv, 802.1AS, 802.1Qcc) offer a strong foundation but require adaptation for dynamic wireless environments.
- Admission control, scheduling, and network planning are central to achieving scalable wireless TSN deployments.
- Promising research includes improved synchronization methods, enhanced traffic management, and security measures to protect time-critical operations.

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Questions?

Link to References